run_output.txt — exported 2025-08-16 01:34

```
Fitting 3 folds for each of 16 candidates, totalling 48 fits
Best C: 0.011659144011798317
CV ROC-AUC (mean): 0.8169151077959099
Validation ROC-AUC: 0.8184
Validation PR-AUC : 0.3630
Validation KS     : 0.5006 @ threshold 0.4859

Validation Confusion Matrix:
[[23954  4038]
 [  712  1293]]

Validation Classification Report:
              precision    recall  f1-score   support

           0     0.9711    0.8557    0.9098     27992
           1     0.2425    0.6449    0.3525      2005

    accuracy                         0.8417     29997
   macro avg     0.6068    0.7503    0.6312     29997
weighted avg     0.9224    0.8417    0.8725     29997


=== TEST RESULTS (using validation KS threshold) ===
Test ROC-AUC: 0.8303
Test PR-AUC : 0.3653
Test KS     : 0.5255

Test Confusion Matrix:
[[24149  3844]
 [  688  1317]]

Test Classification Report:
              precision    recall  f1-score   support

           0     0.9723    0.8627    0.9142     27993
           1     0.2552    0.6569    0.3676      2005

    accuracy                         0.8489     29998
   macro avg     0.6137    0.7598    0.6409     29998
weighted avg     0.9244    0.8489    0.8777     29998

Fitting 3 folds for each of 16 candidates, totalling 48 fits
Best params: {'xgb__learning_rate': 0.05, 'xgb__max_depth': 4, 'xgb__min_child_weight': 3.0, 'xgb__n_estimators': 400}
CV ROC-AUC : 0.8576803590955926
```

```
Validation ROC-AUC: 0.8695
Validation PR-AUC : 0.4082
Validation KS     : 0.5864 @ threshold 0.4632

Validation Confusion Matrix:
[[21692  6300]
 [  378  1627]]

Validation Classification Report:
              precision    recall  f1-score   support

           0     0.9829    0.7749    0.8666     27992
           1     0.2052    0.8115    0.3276      2005

    accuracy                         0.7774     29997
   macro avg     0.5941    0.7932    0.5971     29997
weighted avg     0.9309    0.7774    0.8306     29997


=== TEST RESULTS (using validation KS threshold) ===
Test ROC-AUC: 0.8717
Test PR-AUC : 0.4100
Test KS     : 0.5942

Test Confusion Matrix:
[[21742  6251]
 [  375  1630]]

Test Classification Report:
              precision    recall  f1-score   support

           0     0.9830    0.7767    0.8678     27993
           1     0.2068    0.8130    0.3298      2005

    accuracy                         0.7791     29998
   macro avg     0.5949    0.7948    0.5988     29998
weighted avg     0.9312    0.7791    0.8318     29998

Best CatBoost params: {'depth': 6, 'learning_rate': 0.05, 'l2_leaf_reg': 3.0}
Validation ROC-AUC: 0.8717359662432942
Validation PR-AUC: 0.4126
Validation KS    : 0.5899 @ threshold 0.4953

Validation Confusion Matrix:
```

```
[[22153  5839]
 [  404  1601]]

Validation Classification Report:
              precision    recall  f1-score   support

           0     0.9821    0.7914    0.8765     27992
           1     0.2152    0.7985    0.3390      2005

    accuracy                         0.7919     29997
   macro avg     0.5986    0.7950    0.6078     29997
weighted avg     0.9308    0.7919    0.8406     29997


=== TEST RESULTS (CatBoost, using validation KS threshold) ===
Test ROC-AUC: 0.8618
Test PR-AUC : 0.3919
Test KS     : 0.5715

Test Confusion Matrix:
[[23404  4589]
 [  548  1457]]

Test Classification Report:
              precision    recall  f1-score   support

           0     0.9771    0.8361    0.9011     27993
           1     0.2410    0.7267    0.3619      2005

    accuracy                         0.8288     29998
   macro avg     0.6091    0.7814    0.6315     29998
weighted avg     0.9279    0.8288    0.8651     29998

Fitting 3 folds for each of 16 candidates, totalling 48 fits
Best RF params: {'rf__max_depth': None, 'rf__max_features': 'sqrt', 'rf__min_samples_leaf': 5, 'rf__n_estimators': 800}
CV ROC-AUC    : 0.853579968341507
Validation ROC-AUC: 0.8654
Validation PR-AUC : 0.3950
Validation KS     : 0.5811 @ threshold 0.2539

Validation Confusion Matrix:
[[22703  5289]
 [  461  1544]]
```

```
Validation Classification Report:
              precision    recall  f1-score   support

           0     0.9801    0.8111    0.8876     27992
           1     0.2260    0.7701    0.3494      2005

    accuracy                         0.8083     29997
   macro avg     0.6030    0.7906    0.6185     29997
weighted avg     0.9297    0.8083    0.8516     29997


=== TEST RESULTS (Random Forest, using validation KS threshold) ===
Test ROC-AUC: 0.8634
Test PR-AUC : 0.3923
Test KS     : 0.5761

Test Confusion Matrix:
[[22746  5247]
 [  476  1529]]

Test Classification Report:
              precision    recall  f1-score   support

           0     0.9795    0.8126    0.8883     27993
           1     0.2256    0.7626    0.3483      2005

    accuracy                         0.8092     29998
   macro avg     0.6026    0.7876    0.6183     29998
weighted avg     0.9291    0.8092    0.8522     29998


=== TEST summary ===
       Model  ROC-AUC   PR-AUC       KS  KS_threshold
     XGBoost 0.871716 0.409956 0.594207      0.455568
RandomForest 0.863386 0.392263 0.576084      0.255425
    CatBoost 0.861847 0.391867 0.571462      0.407810
    L1-Logit 0.830288 0.365291 0.525482      0.455505

=== TEST Portfolio VaR & ES (counts and rates; assumes indep. defaults, EAD=1, LGD=1) ===
       Model     MeanLoss   StdLoss  VaR@95        ES@95  VaR@97.5      ES@97.5  VaR@99        ES@99  MeanLossRate  VaRrate@95  ESrate@95
VaRrate@99  ESrate@99
    L1-Logit 11203.885625 77.528844 11331.0 11362.782716   11357.0 11384.113300 11382.0 11408.518072      0.373488    0.377725   0.378785
0.379425   0.380309
     XGBoost  9444.348375 66.080618  9555.0  9581.977556    9575.0  9600.158416  9598.0  9622.225000      0.314833    0.318521   0.319421
```

```
0.319955   0.320762
    CatBoost  8274.475625 63.225694  8378.0  8403.397500    8396.0  8419.846535  8421.0  8439.150000      0.275834    0.279285   0.280132
0.280719   0.281324
RandomForest  5244.216250 55.603381  5336.0  5359.689487    5354.0  5376.450495  5374.0  5395.423529      0.174819    0.177879   0.178668
0.179145   0.179859

Saved PDF: run_output.pdf
```